

SOFTWARE ENGINEERING LAB 4

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PROMPT 1:

So, I am a gamer who wants to make the following changes given below

Task 1 — Guarantee solvable starting states

Change puzzle initialisation so every generated board is reachable from the solved arrangement. Prefer generating the solved board and applying legal blank moves to scramble it.

Done when: repeated fresh puzzles are solvable in principle and the puzzle never starts in an unreachable parity state.

Task 2 — Complete puzzle lifecycle

Detect the solved arrangement and end the game cleanly. Prevent commands from changing state after completion.

Task 3 — Size, timer, and move modes

Add 3x3, 4x4, and 5x5 modes. Track moves and elapsed time consistently across all sizes. Do not reset session values accidentally when only the board is recreated.

Task 4 — Valid-action feedback

Only report a successful slide when a tile actually moved into the blank. Invalid moves must not increment the move count or generate success feedback.

Required testing

Generate many fresh boards, test all sizes, solve a small board, attempt impossible moves, verify move counts, verify timer behaviour, test invalid commands, and quit.

so can you help me make these changes and also highlight the changes you are making while explaining in short how you are doing